

AI-POWERED CAD ANALYSIS REPORT

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Geometry to Process to Cost · Fully AI-reasoned

49

Cast Suspension/Structural Arm (large organic bracket, sand-cast)

58 × 274 × 115 mm · 429.4 cm³ · Al 1.159 kg / Steel 3.371 kg

Manufacturability: 49/100 · Confidence: Low

§1 — Geometry & Part Summary

Bounding Box	58.2 × 274.5 × 115.0 mm	Surface Area	829.8 cm ²
Volume	429.43 cm ³	Weight (Al)	1.159 kg
Weight (Steel)	3.371 kg	Weight (Plastic)	0.451 kg

§2 — Detected Features

Feature Type	Count	Significance	Description
Holes	15	High	15 through/blind holes (r 4-25mm) requiring drilling/boring post-cast
Threaded features	4	Medium	Tapped/threaded bores detected, requiring tapping or thread-milling secondary op
Undercuts	44	High	44 undercut faces relative to Z draw direction — needs side-cores/slides or re-orientation
Zero-draft faces	68	High	68 faces with 0° draft — high risk of pattern/die sticking and drag on ejection
Free-form (B-spline/torus) surfaces	104	Medium	62 B-spline + 42 torus faces indicate an organic, blended/filletted casting shape typical of a structural or suspension arm
Large-radius cylindrical surfaces	5	Low	Cylindrical faces up to r=222mm suggest a large swept boss/bearing seat or curved rib

§3 — Material Analysis

AI-suggested

Primary Material: Ductile (nodular) cast iron, GJS500 (55% confidence)

No aluminium signal exists: mean wall 25.8mm / max 49.75mm is far too thick for HPDC aluminium (2-4mm typical) and the part shows classic organic, undercut-heavy, thick-section geometry of a structural/suspension arm or bracket. Per guidance, thick-section drivetrain/suspension-type solids default to steel/cast iron rather than aluminium. Ductile iron GJS500 is the standard low-cost, high-strength material for sand-cast automotive structural arms at high volume, and matches the deterministic cast-iron weight (3.070kg) provided.

Alternatives: Austempered Ductile Iron (ADI) (20%) · LM25 Aluminium casting alloy (if lightweighting confirmed) (15%) · Higher-strength ductile iron G

§4 — Process Recommendations

Process	Commodity	Confidence	Cycle Time (hr)	Reasoning
Sand Casting (automated/high-pressure moulding line)	casting	78%	0.2730	Thick, non-uniform walls, 44 undercuts, and organic B-spline/torus surfaces are best produced by sand casting with cores for undercuts; high annual volume (200k) is achievable on automated flaskless sand-moulding lines for ductile iron.
Cast + Machine (secondary drilling/tapping/finish)	cast_and_machine	15%	0.6090	15 holes and detected threads require post-cast drilling, boring and tapping; 3-setup CNC operation follows the sand-cast rough part.
Closed-die Forging	forging	7%	0.1500	Forge die cost (£62,000) and stroke count (8) were provided as a parametric alternative; forging could be used for a higher-strength steel version of this arm, but the freeform/undercut geometry favours casting.

§5 — Manufacturability Risks (Score: 49/100)

Severity	Feature / Area	Description	Recommended Action
High	Undercuts (44 faces)	Large number of undercut faces relative to the Z draw direction will require multiple side-cores or slides in the pattern/die, increasing tooling cost and cycle time.	Re-orient part or split undercut features into cored sections; consolidate undercuts to minimise number of slides.
High	Zero-draft faces (68)	68 faces have 0° draft, risking pattern sticking, drag defects, and increased scrap in sand moulding.	Add minimum 1-3° draft to all vertical faces not functionally critical.
Medium	Wall thickness variation (Δ = 11.67 mm)	Wide variation from 6.19mm to 49.75mm creates differential cooling rates, risking shrinkage porosity and hot-tearing at the 49.75mm sections.	Add fillets/tapered transitions and consider local chills or risers at thick sections.
Medium	15 holes + threads	Significant post-cast machining load (drilling, boring, tapping) adds cost and cycle time beyond the raw casting.	Cast some holes near-net with cores to reduce machining stock, and cluster machining features to minimise setups.
Low	Free-form surfaces (62 B-spline, 42 torus)	High organic surface content increases pattern-making/tooling machining complexity and cost.	Simplify blends where not functionally required to ease pattern manufacture.

§7 — Cost Range & Suggested Inputs

OPTIMISTIC	MOST LIKELY	CONSERVATIVE
£0.00	£0.00	£0.00

Net Weight	3.070 kg	Material	mat-gjs500
Recommended Process	casting	Cycle Time	—
Setup Time	—	Operations	0 ops

Operations Detail

Process-Specific Parameters

Casting Subtype	sand
Die/Mould Cost	£15,000
Die Life	8,000 shots
Cavities	—
Yield	65.0%

§8 — AI Analysis Explanation